

Probabilistic Graphical Models 1 - Representations

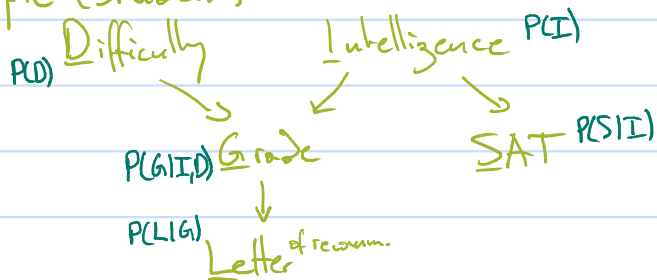
A **factor** with **scope** $\{X_1, \dots, X_k\}$ is a map $\phi: \text{Val}(X_1, \dots, X_k) \rightarrow \mathbb{R}$.

Example: The joint pdf of $\{X_1, \dots, X_k\}$ conditioned on $\{X'_1, \dots, X'_l\}$

Factors can be **multiplied**, with the union scope as a scope, or **marginalised** wrt a subset of the scope (by integrating along the subset), or **reduced**, by restricting a subset of the scope to single values.

Bayesian networks are directed acyclic graphs (DAGs) with random variables as vertices, and pdfs $P(X_i | \text{Par}(X_i))$. They show dependencies between probability distributions.

Example (Student):



e.g. $P(D, I, G, S, L) = P(L|D, I, G, S) P(D, I, G, S)$, etc.

$$\leadsto P(D, I, G, S, L) = P(D) P(I) P(G|I, D) P(L|G) P(S|I)$$

A pdf P **factorises over a DAG G** if $P(X_1, \dots, X_n) = \prod P(X_i | \text{Par}_G(X_i))$.

A **trail** $X_1 - \dots - X_n$ is a path in the undirected DAG. A trail is **active** if it does not contain any **v-structures** $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$.

Note that X_1 influences X_n iff there is an active trail between them.

In all the following connections, X can influence Y :

$$X \rightarrow Y, X \leftarrow Y, X \rightarrow W \rightarrow Y, X \leftarrow W \leftarrow Y, X \leftarrow W \rightarrow Y$$

Note that conditioning on W above blocks the influence, while it makes the v-structure $X \rightarrow W \leftarrow Y$ active! Note also that conditioning on any descendant of W makes it active!

In the case of conditioning w.r.t. a subset Z , the trail is **active** iff for any of its v-structures $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$, X_i or one of its descendants is in Z , and no other X_j is in Z .

For two events α, β and a probability distribution P we say that $P \models \alpha \perp \beta$ if $P(\alpha, \beta) = P(\alpha)P(\beta)$, and similarly for random variables X, Y .

X and Y are **d-separated** given Z if there is no active trail in G given Z . We say that $\text{d-sep}_G(X, Y | Z)$.

Theorem: If P factorises over G then $\text{d-sep}_G(X, Y | Z) \Rightarrow P \models X \perp Y | Z$.

Note: Any node is d-separated from its non-descendants given its parents.

Let $I(G) = \{(X \perp Y | Z) : \text{d-sep}_G(X, Y | Z)\}$. If P satisfies $I(G)$ then G is called an **I-map** (or **independency map**) of P .

$$I(D \perp I) = \{D \perp I\}, I(D \rightarrow I) = \emptyset$$

satisfied by all P

Theorem: P factorises over G iff G is an I-map of P .

Example (Naive Bayes model): Consider In particular, $X_i \perp X_j | C$, and $P(C, X_1, \dots, X_n) = P(C) \prod P(X_i | C)$.

$$\text{hence } \frac{P(C=c^1 | x_1, \dots, x_n)}{P(C=c^2 | x_1, \dots, x_n)} = \frac{P(C=c^1)}{P(C=c^2)} \prod \frac{P(x_i | C=c^1)}{P(x_i | C=c^2)}$$

For the Bernoulli Naive Bayes model, we have i.e. the features are Bernoulli variables. For the

Multinomial NBM, we have We could additionally assume that the w_i are identically distributed.

Temporal Models

Let $X^{(t)}$ denote the variable X at time t , $X^{(t:t')} = \{X^{(t)}, \dots, X^{(t')}\}$. We say that X is **Markov** if $\forall t > 0$, $(X^{(t+1)} \perp X^{(0:t-1)} | X^{(t)})$, i.e. t is discrete.

$$P(X^{(0:T)}) = P(X^{(0)}) \prod_{t=1}^T P(X^{(t)} | X^{(0:t-1)})$$

$$= P(X^{(0)}) \prod_{t=1}^T P(X^{(t)} | X^{(t-1)}).$$

A **2-time slice Bayesian network** (2TBN) over $X_1, \dots, X_n$ is a Bayesian network fragment with nodes including $X_1^{(t)}, \dots, X_n^{(t)}$ and a subset of $X_1, \dots, X_n$, where only the $X_i^{(t)}$ have parents and a conditional (on the X_i) pdf.

Note that a 2TBN defines a conditional pdf

$$P(X' | X) = \prod P(X_i' | \text{Par } X_i').$$

A **dynamic Bayesian network** (DBN) is a 2TBN $\text{BN} \rightarrow$ over $X_1, \dots, X_n$ together with a Bayesian network $\text{BN}^{(0)}$ over $X_1^{(0)}, \dots, X_n^{(0)}$. We define a **ground** (or **unrolled**) **network** by using the 2TBN to inductively create $\text{BN}^{(t+1)}$.

A **hidden Markov model** (HMM) is a DBN in which the 2TBN is a Markov process.

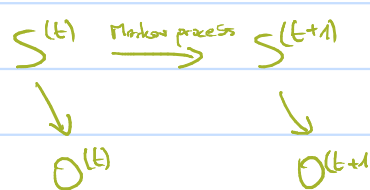
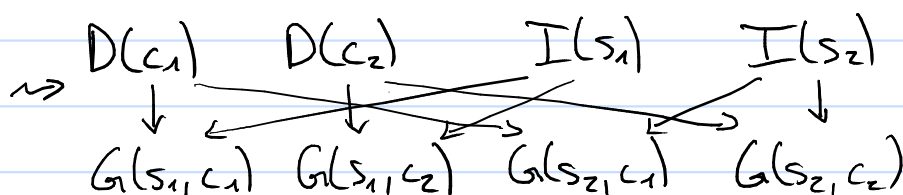
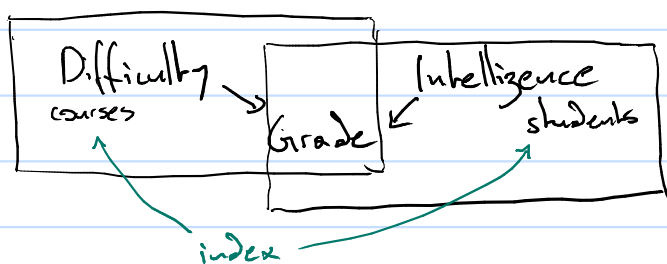
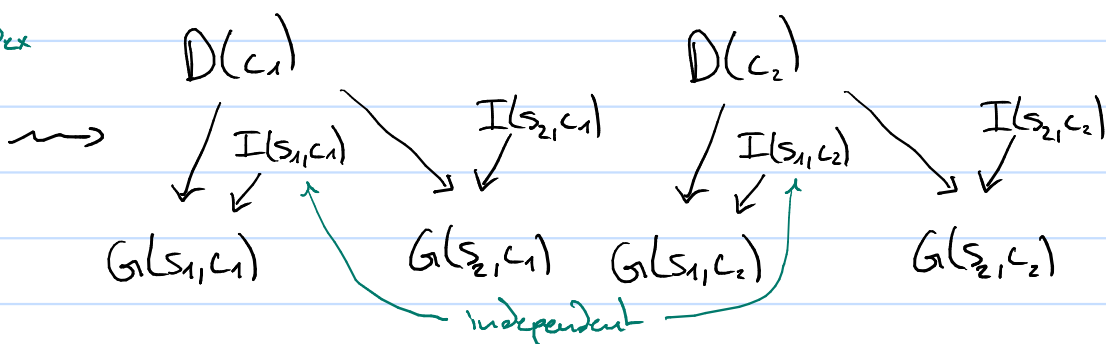
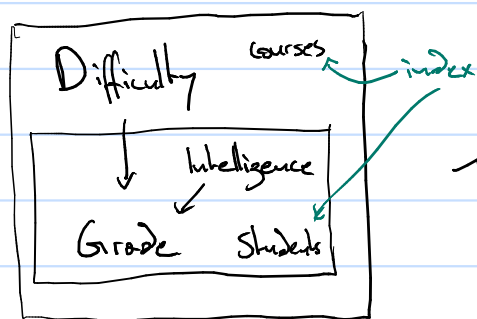


Plate Models



A **plate dependency model** is a collection of template variables A indexed by $U_1, \dots, U_k$ and template parents $B_1, \dots, B_k$ indexed by $U_1, \dots, U_k$, respectively, together with a conditional pdf $P(A | B_1, \dots, B_k)$. We get a ground network by instantiating each U_i .

Structured conditional pdfs

i.e. $P(X, Y | C = c) = P(X | C = c) P(Y | C = c)$

Let c be an assignment to C . We say that X, Y are **context-specifically independent** w.r.t. c if $P(X \perp_c Y | Z, c)$.

Let Y_1, Y_2 . Then $X \perp Y_1 | Y_2^0$ and even $Y_1 \perp Y_2 | x^0$.

$X = Y_1 \vee Y_2$

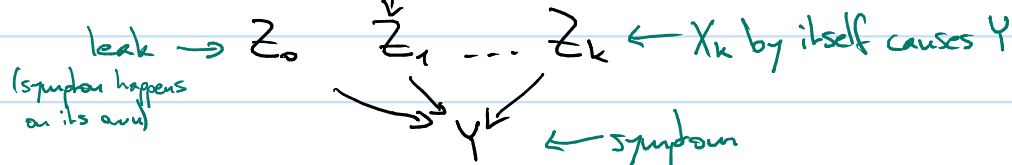
Consider the network $Z_1 \dots Z_k$. Then

$\{1, \dots, k\} \ni A \rightarrow Y = Z_A$
 $\uparrow$ select Z_A for Y

$$P(Y | A, Z_1, \dots, Z_k) = \begin{cases} 1 & \text{if } A = a, Z_a = Y_a \\ 0 & \text{o/w} \end{cases}$$

is called a **multiplexer conditional pdf**. It is an example of a **tree-structured conditional pdf** which can be written as a decision tree of its parent variables.

Consider the network $X_1 \dots X_k \leftarrow \text{diseases}$

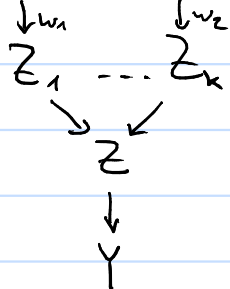


with $P(Z_i = 1 | X_i) = \begin{cases} 0 & \text{if } X_i = 0 \\ \lambda_i & \text{if } X_i = 1 \end{cases}$ and $Y = Z_0 \vee Z_1 \vee \dots \vee Z_k$. Then

$$P(Y = 0 | X_1, \dots, X_k) = (1 - \lambda_0) \prod_{i: X_i = 1} (1 - \lambda_i)$$

is called the **noisy OR conditional pdf**. Note that $(X_1 \perp_c X_2 | Y^0)$.

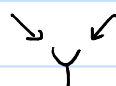
Consider the network $X_1 \dots X_k$ where $Z_i = w_i X_i$ and $Z = w_0 + \sum w_i X_i$. Then



$z \mapsto \frac{e^z}{1 + e^z}$

the **sigmoid cond. pdf** is $P(Y^1 | X_1, \dots, X_k) = \text{sigmoid}(Z)$.

Consider the DAG $X_1 \dots X_k$. Then Y is a **linear Gaussian** if



$Y \sim \mathcal{N}(w_0 + \sum w_i X_i; \sigma^2)$
 $\uparrow$ fixed

If Y is dependent on an additional discrete variable A and is linear Gaussian conditioned on any $A = a$ then it is called **conditional linear Gaussian**.

Markov Networks

Consider the graph $X_4 - X_1 - X_2$ of binary variables. Let $\phi_i(X_i, X_{i+1})$ be some function, $P(X_1, \dots, X_4) = \frac{1}{Z} \phi_1(X_1, X_2) \dots \phi_4(X_4, X_1)$.
 (Note: ϕ_i is an affinity function, and the product is mod 4. The $\frac{1}{Z}$ is a normalise factor.)

A pairwise Markov network is an undirected graph with nodes $X_1, \dots, X_k$ and with a factor $\phi_{ij}(X_i, X_j)$ associated to each edge $X_i - X_j$.

Let D_i be sets of random variables, $\phi_i(D_i)$ factors, $\Phi = \{\phi_i(D_i)\}$. Then the Gibbs distribution is $P_\Phi(X_1, \dots, X_n) = \frac{1}{Z_\Phi} \prod \phi_i(D_i)$. It induces a Markov network H_Φ with $X_i - X_j$ if $X_i - X_j \in D_k$ for some k .

We say that P factorises over H if there is $\Phi = \{\phi_i(D_i)\}$ st. $H = H_\Phi$ and $P = P_\Phi$.

A trail $X_1 - \dots - X_n$ is active given Z if no X_i is in Z .

Let $\Phi = \{\phi_i(D_i)\}$, $\tilde{P}_\Phi = \prod \phi_i(D_i)$, $Z_\Phi(X) = \sum_Y \tilde{P}_\Phi(X, Y)$. Then $P_\Phi(Y|X) = \frac{1}{Z_\Phi} \tilde{P}_\Phi(X, Y)$

is called a conditional random field parameterised by X .

For the logistic model, $\phi_i(X_i, Y) = \exp(w_i I(X_i = Y = 1))$. Then $\phi_i(X_i, Y = 1) = \exp(w_i X_i)$, $\phi_i(X_i, Y = 0) = 1$. Hence, $\tilde{P}_\Phi(X, Y = 1) = \exp(\sum w_i X_i)$, $\tilde{P}_\Phi(X, Y = 0) = 1$, and

$$P_\Phi(Y = 1 | X) = \frac{\exp(\sum w_i X_i)}{1 + \exp(\sum w_i X_i)}$$

We say that X, Y are separated in H given Z if there is no active trail between them, given Z .

Theorem: If P factorises over H and $\text{sep}_H(X, Y | Z)$ then $P \models (X \perp Y | Z)$.

We similarly define $I(H)$ and I -maps for P , and get the same results as for Bayesian networks. Here, we also get the converse:

Theorem (Hammersley-Clifford): Let P be a positive distribution. If H is an I -map for P then P factorises over H .

We will write $I(P) = \{(X \perp Y | Z) : P \models (X \perp Y | Z)\}$. Note that if P factorises over G then $I(G) \subseteq I(P)$.

A minimal I -map for P is a graph G in which $I(G)$ changes whenever an edge is removed.

Note that different such G may have different numbers of edges:

$$I\left(\begin{array}{ccc} D & & I \\ & \searrow & \nearrow \\ & G & \end{array}\right) = I\left(\begin{array}{ccc} D & \rightarrow & I \\ & \searrow & \nearrow \\ & G & \end{array}\right)$$

A perfect map for P is a graph G s.t. $I(G) = I(P)$.

Note that in the example above of $P = P_\Phi$ for $D \rightarrow A \rightarrow B$ we have $P \models \{A \perp\!\!\!\perp C | B, D\}, (B \perp D | A, C)\}$,

but P does not have a perfect map.

Note that neither does $X_1 - X_2$. Here, $X_1 \perp\!\!\!\perp X_2, X_1 \perp\!\!\!\perp Y, X_2 \perp\!\!\!\perp Y$.

$$\begin{array}{ccc} X_1 & & X_2 \\ & \searrow & \nearrow \\ & Y & \end{array} \quad Y = X_1 \oplus X_2$$

Similarly, we define perfect maps using Markov networks.

Note that perfect graphs are not unique:

$$\begin{array}{ccc} X & \rightarrow & Y & \rightarrow & Z \\ X & \leftarrow & Y & \rightarrow & Z \\ X & \leftarrow & Y & \leftarrow & Z \end{array} \quad \text{have } I(G) = \{X \perp\!\!\!\perp Z | Y\}.$$

Two graphs G_1, G_2 are I -equivalent if $I(G_1) = I(G_2)$.

In a log-linear model we replace $\tilde{P} = \prod \phi_i(D_i)$ by

$$\tilde{P} = \exp\left(-\sum w_j f_j(D_j)\right).$$

(features may share scopes)

The Ising model is a pairwise Markov network with binary X_i and

$$P(X) \propto e^{-\frac{1}{T} E(X)}$$

($\uparrow$ in $\{ \pm 1 \}$)

where $E(x_1, \dots, x_n) = -\sum_{\substack{i,j \\ x_i = x_j}} w_{ij} x_i x_j - \sum u_k x_k$.

($\leftarrow$ given a graph)

Let X_i have values in a metric space (V, μ) of labels. We get a metric Markov

random field by setting $\tilde{P}(X) = \exp(-\sum w_{ij} \mu(X_i, X_j))$.

Decision making

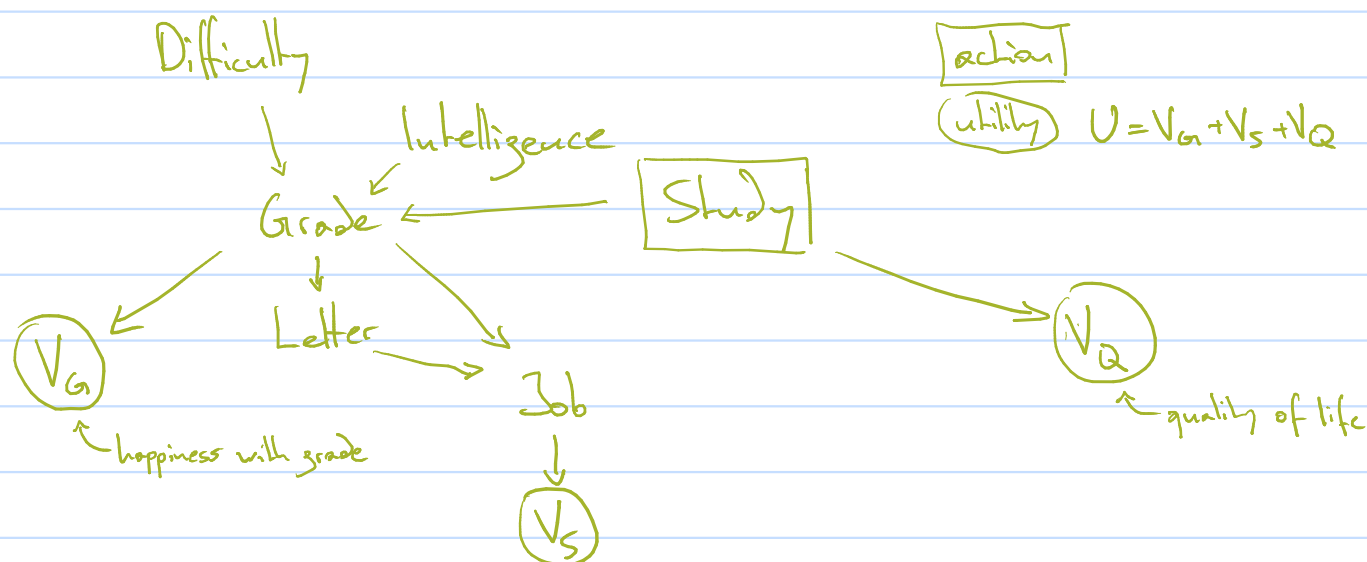
A **decision situation** $\mathcal{D}$ consists of

- a set of actions $\text{Val}(A) = \{a^1, \dots, a^k\}$,
- a set of states $\text{Val}(X) = \{x^1, \dots, x^n\}$,
- a distribution $P(X|A)$,
- a utility function $U(X, A)$.

The **expected utility** of an action a is $EU(x) = \sum_x P(x|a) U(x, a)$.

We want to find $a^* = \arg\max_a EU(x)$.

An **influence diagram** is a Bayesian network together with a set of actions and a utility function.



A **decision rule** S at an action node A is a conditional pdf $P(A|\text{Par } A)$. The edges between A and $\text{Par } A$ are **information edges**.

We could add a decision rule above for $\text{Difficulty} \rightarrow \text{Study}$.

The **expected utility with information** is $EU(S_A) = \sum_{x,a} P_{S_A}(x,a) U(x,a)$,
prob. dist. associated with $\bar{X} \cup \bar{A}$

and we now want to maximise $EU(S_A)$.

Note that

$$\begin{aligned} EU(S_A) &= \sum_{x_1, \dots, x_n, A} \left(\left(\prod P(x_i | \text{Par } x_i) \right) U(\text{Par } U) S_A(A|Z) \right) \\ &= \sum_{Z, A} S_A(A|Z) \sum_x \left(\left(\prod P(x_i | \text{Par } x_i) \right) U(\text{Par } U) \right) \\ &= \sum_{Z, A} S_A(A|Z) \mu(A, Z). \end{aligned}$$

parents of utility nodes
Par A, observations prior to A

Then $S_A^*(a, z) = \begin{cases} 1 & \text{if } a = \arg\max_A \mu(A, z) \\ 0 & \text{o/w} \end{cases}$.

The **value of perfect information** about X , $VPI(A|X)$, is the value of observing X before choosing an action in A :

$$VPI(A|X) = \text{MEU}(\mathcal{D}_{X \rightarrow A}) - \text{MEU}(\mathcal{D}).$$

influence diagram with additional edges $X \rightarrow A$
max. exp. utility

information is useful iff it changes my decision

Theorem: $VPI(A|X) \geq 0$ with equality iff the optimal decision rule for $\mathcal{D}$ is optimal for $\mathcal{D}_{X \rightarrow A}$.

Knowledge Engineering

Types of models

Template based

image segmentation

fault diagnosis

small number of variable types

Specific

medical diagnosis

features are predictive

Generative

task shifts

easier to train when labels

are not fully labelled

Discriminative

particular predictive task

Variable types

Target

Observed

Latent

hidden

time

watch1

watch2

Structure

Causal vs. non-causal

Note that any model on $X \rightarrow Y$ is also modelled on $Y \rightarrow X$. Causal models are often sparser.

Local structure of parameters

Discrete

Continuous

Context specific

tree cond. pdf

regression tree

↑ with thresholds

Aggregating

sigmoid, noisy OR

linear Gaussian